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MyDesign Portfolio

Element G Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your construction of a testable prototype, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own prototype construction details, be sure to delete the instructions.

Guidance before filling out the template that follows:

- Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of this Element as another piece of a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element also.
- You will then proceed on to test and analyze this design and then determine your next iteration, returning to this Element afterwards when you are ready to construct a new testable prototype. Fill out each of these steps again.

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

This section focuses on the process of explaining your prototypes. While the title of the subelement focuses on your final prototype, you must have at least an initial prototype and a final prototype, with additional prototypes in-between being desirable. Note that you never know which prototype is your final prototype until after you test it, so be sure to record all of the information for each prototype. Use as many copies of this sub-template as needed.

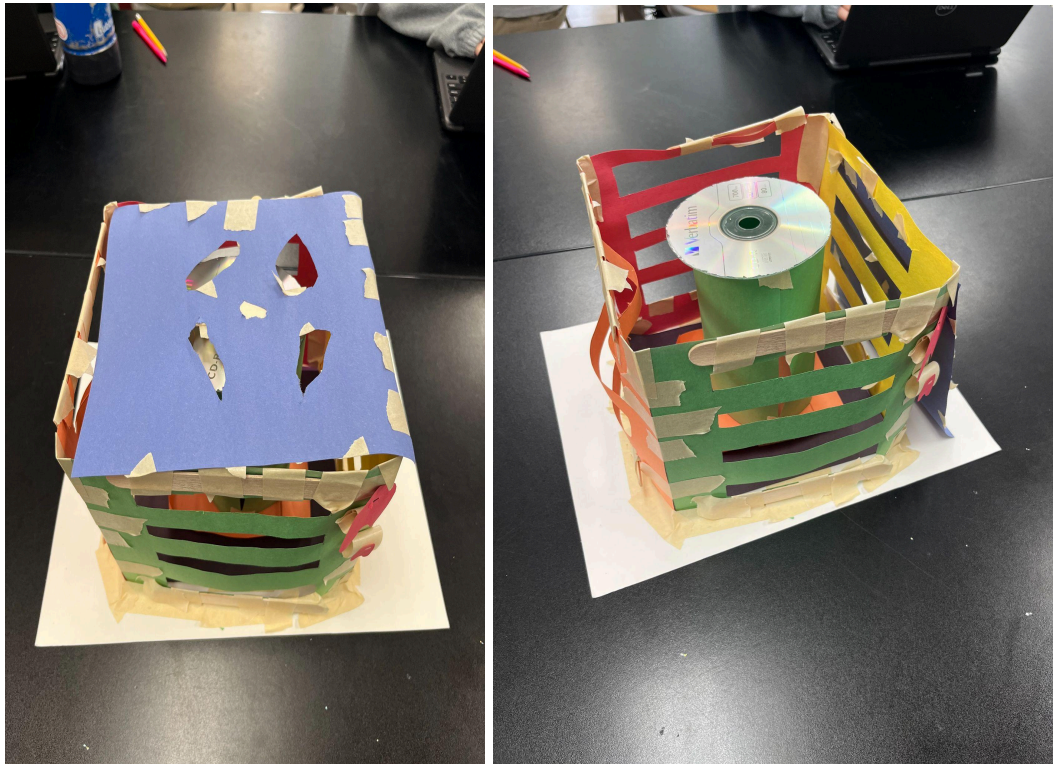
Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

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Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)



	Details	Tools and Manufacturing Methods used
Step 1	Get six construction pieces of paper.	Construction paper
Step 2	Cut 4 rectangle slips into four of the six pieces of paper. Then cut a circular shape into one piece of the paper, this	Exacto knife

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	one will represent the top.	
Step 3	Using tape to assemble the six pieces of paper together. The blank piece of paper should be the bottom, the rectangular cuts should be the four sides, and the circular one is a top.	tape
Step 4	Using an old disc cut and another sheet of construction paper, roll the construction paper up and tape the CD on top of the roll. This will represent the air filter.	CD, Scissors, Construction paper, tape
Step 5	Tape the circular "air filter" on the inside of the paper box.	tape
Step 6	To create a more sturdy structure, we taped popsicle sticks and tape to the sides.	Popsicle sticks, tape



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Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #2 (change this number to match the design number you're on)



	Details	Tools and Manufacturing Methods used
Step 1	Cut a circular hole on the top of the plastic box. You'll need to use a saw and dremel, a dremel will help smooth the sides.	Dremel and saw

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Step 2	Cut rectangle slits into the sides of the plastic box. Using the same materials, a dremel and a saw.	Dremel and saw
Step 3	Assemble the fan and screw it onto the top of the plastic box. The fan should be screwed on top of the hole in the plastic box. The sides should be taped in order to recreate a vacuum tight seal.	Screwdriver and screws, duct tape
Step 4	Create a mold to secure the filter into a position, we upcycled a cardboard box. Trace around the circular filter and cut out the shape of the filter. Place the mold under the fan to push air through the filter.	Cardboard and exacto knife
Step 5	Put a filter into the cardboard mold in the box so air can be filtered.	Filter

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Step 6	Close the box and add stickers to the outside.	Stickers
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Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
Fan	Cooling fan	Amazon	\$13.99	1	\$13.99
Box	Hinged plastic box	Amazon	\$44.00	1	\$44.00
Filter	Pool filter	Amazon	\$25.00	1	\$25.00
Stickers	Stickers	Amazon	\$7.99	1	\$7.99
Screws	Wood screws	Amazon	\$7.59	1	\$7.59
USB extender cord	10 foot USB cord	amazon	\$7.99	1	\$7.99
USB plug-in	2 USB plug in	Amazon	\$8.00	1	\$8.00

Detailed sketches, which might include CAD if you have learned it in your class:

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Construction of the Final Prototype (G2)

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Portable	Our prototype is not big and it's effective and at a portable size at the same time.
Plug in	It is plugged in because it's easier so that you don't have to change your batteries.
Small rectangle strips	Cut out the sides of the box so that the air would come through and let the filtering happen easier.
Light-weight	The box used to house the filter and fan is meant to be reasonable in weight. This ensures that we can hang the box, and is easy to move. This corresponds with Portability.
Filter air	The fan and the filter will remove harmful particulates from the air, this can be measured with Dr Jacksons air quality meter .
The speed air is filtered	The fan will push contaminated air into the filter, we can time how long it takes to get to a safe level of air contamination.

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Quiet	The fan may make noise when active. We already know this information from the website we bought the fan on.
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Testing or Modeling of Attributes (*G3, G4*) and Addressing Non-Testable Attributes (*G4*)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter "expert review required" for these attributes in the final column.



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This is the starting point, it's green.
PM2.5→ 001 mg/m3 (Green good)
TVOC→ 0.029 mg/m3 (Green Good)
HCHO→ 0.011 mg/m3 (Green good)
CO2→0464 ppm (green good)



This is toward our highest peak.
PM2.5→ 860 mg/m3 (red bad)
TVOC→ 0.089 mg/m3 (Green Good)
HCHO→ 0.016 mg/m3 (Green good)
CO2→0584 ppm (green good)

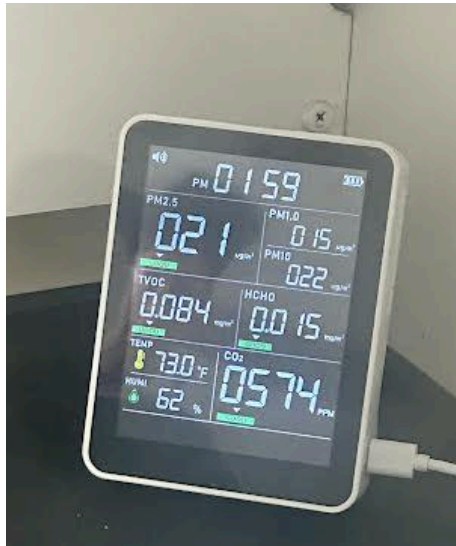


Highest peak, it's in the red.
PM2.5→ 999 mg/m3 (red bad)
TVOC→ 0.120 mg/m3 (Green Good)
HCHO→ 0.023 mg/m3 (Green good)
CO2→0661 ppm (green good)

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The final numbers and scores.

It's in the green.

PM2.5→ 021 mg/m³ (Green good)

TVOC→ 0.084 mg/m³ (Green Good)

HCHO→ 0.015 mg/m³ (Green good)

CO2→0574 ppm (green good)

After 6 minutes

Attribute	Description	Describe your prototype's functionality to test this attribute
Filtering power for particulate	The ability to filter particulate out of the air	The tests showed the device lowering the ppm of particulate matter in the air from 999 ppm to 21 ppm
Filtering power for VOCs	The ability to filter VOCs from the air	The test showed the device lowering the VOC level from 0.120mg/m ² to 0.084 mg/m ²
Low Noise production	The ability to run the device at a low volume	This was tested during the design-a-thon presentation when our stockholders could not hear the fan when it was running on its highest output



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Addressing Non-Testable Attributes (G4)

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Durability	If we were to test durability we may break the prototype in the process and thus waste our budget money.
Visually appealing	How visually appealing something is is a subjective matter and is different from person to person.
The size of the filter	We can't change the size of the filter without altering the size of the box. The same goes with the box, filter, and fan. All of the items are measured and are meant to fix together.
Battery power	We are unable to switch to a battery filtered air purifier, due to the fan being a plug in fan.

